Excel + AI: Hands-on Assignment

Data Analytics Bootcamp 6.0 | Excel + AI Chapter

You watched the assistant do it. Now you drive. Same movie table you saw in the videos, but **new rules**, so the prompts from the videos will not get you there. You will have to say the rule yourself.

This is for your own practice. Nothing to submit.

What you need

- `Assignment_Movies_Workbook.xlsx`
- Any one assistant from the setup guide: Claude in Excel, Gemini in Google Sheets, or free ChatGPT

The ground rules

- **Write your own prompts.** Copying the video prompts will not work, because the rules below are different.
- **Use live formulas, not typed-in values.** If you change a budget, your new columns should update.
- **Check before you accept.** Every task ends with something you verify yourself. That part is the real assignment.

Task 1: Understand the workbook before you touch it

Ask the assistant to walk you through `MovieFinancials` like you just inherited it.

Find out and write down:

1. How `budget USD` is worked out from `budget`, `unit` and `currency`. Which columns are typed in and which are derived?
2. At least **three data quality problems** in the table. Hint: look closely at titles, ratings, studios and the money columns.

Write it here: add a sheet called `My Notes` and put your answers in plain sentences.

Task 2: Turn two business rules into formulas

Add these two columns to the `MovieFinancials` table.

Rule A: Watch Verdict

Verdict	When
Must Watch	IMDb rating is 8 or above
Worth a Watch	Rating is 6 or above, but below 8
Skip	Rating is below 6
Not Rated	Rating is missing or is not a number

Rule B: ROI %

- ROI % = (revenue USD minus budget USD) divided by budget USD
- Show it as a percentage with no decimals

- If budget or revenue is missing, leave the cell blank. No errors on the sheet.
- Conditional formatting: highlight **loss-making movies in red** and any movie with **ROI above 1000% in amber**

Check it yourself:

- Pick a movie rated exactly **8.0**. Did it land in Must Watch? If not, your rule was read differently from what you meant.
- Find the movie with the "NULL" rating. What did it get?
- Pick the movie with the highest ROI. Work out its ROI on a calculator. Does it match?

Task 3: One summary, one chart

On a new sheet called `Language Summary`, build a summary by **language name**, not language id. You will need the `languages` sheet for that.

For each language show:

- Number of movies
- Average IMDb rating
- Total profit in USD millions (revenue USD minus budget USD)

Then chart **total profit by language**.

One more rule: **show every language in the languages sheet, including ones with no movies.** Those rows should show 0 movies, not vanish.

Check it yourself:

- Does your movie count add up to the total number of rows in `MovieFinancials`?
- Does your average rating quietly treat "NULL" as zero? Recalculate one language by hand.
- Look at the chart. One language dwarfs the rest. Is that chart useful to a director as it is, or would you change it? Write one line on that.

Task 4: Run the five checks on your own work

This is the habit the last video was about. Add a sheet called `Check Log` and fill this in:

Check	What I did	Result
1. Row count		
2. One value by hand		
3. Grand total vs sum of the parts		
4. Filter stress test		
5. Assumptions the AI made that I did not ask for		

For check 5, read back what the assistant **said** it did, not just what it produced. Name at least one choice it made on its own.

You are done when your workbook has

1. `MovieFinancials` with **Watch Verdict** and **ROI %** columns and the colour rules
2. `My Notes` (Task 1)
3. `Language Summary` with the chart (Task 3)
4. `Check Log` (Task 4)
5. A sheet called `Prompts`: paste every prompt you used, in order. Mark one where the first answer was wrong and note how you fixed your prompt.

A perfect-looking sheet with an empty Check Log means you skipped the real exercise. The rule stays yours, and so does the checking.

One rule you never break

This file is practice data, so use any tool you like. At work, **never paste client data into a public AI tool.**